

Muhammad Ali Farooq (he/ him)

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Career Objectives

A dynamic and results-oriented professional with extensive experience in both research and development and project management. Possessing advanced expertise in the domains of automation, generative AI, computer vision, machine learning, machine vision, medical imaging, edge computing, circuit design and sensor fusion algorithms, with a keen focus on achieving tangible business objectives and academic research goals.

Education

September 2019 - August 2022, Ph.D., Electrical and Electronic Engineering, National University of Ireland Galway

Grade: Successful Ph.D. remote defence

Thesis: [Evaluations of Thermal Imaging Technology for Automotive Use Cases](#)

Other Projects: Medical image analysis, Digital image processing, and enhancement, Facial depth map estimations, 3D face construction using deep learning.

Subjects: Research Integrity

August 2014 - May 2017, Master of Science (MS), Electrical Control Engineering, National University of Sciences and Technology Pakistan

CGPA: 3.50 on a scale of 4.00

Thesis: [Automatic Lesion Detection System for Skin Cancer Classification using SVM and Neural Classifiers](#)

Other Projects: PID Controller design for Industrial Automated Control Systems.

Subjects: Computer Vision, Advanced Control System, Advanced Embedded Systems design, Advanced Digital Signal Processing, Digital Data Acquisition, Probability, and Statistics

August 2008 - December 2012, Bachelor of Engineering (BE), Electronic Engineering, IQRA University

CGPA: 3.24 on a scale of 4.00

Final Year Project: [Advanced Digital Incubation System using Arduino Controller](#)

Other Projects: Smart Grid, Switch Mode Power Supply, PIC Micro-Controller based Home Automation System

Subjects: Digital Electronics, Power Electronics, Digital Logic Design, Industrial Electronics, Applied Mathematics, Circuit Analysis, and Digital Communication Systems

Work Experience

October 2024 – Currently Working as a Research Fellow, at University of Galway and Machine Learning Researcher, at FotoNation Galway

- Leading multiple large-scale EU and SFI funded projects including Prive-Sense (Privacy-Responsive Intelligent Safe sENSing and Situational Enhancement), DTIF and Adapt project. The project focuses on building intelligent AI perception pipelines by the combinations of multimodal imaging including near-Infrared (NIR), thermal-Infrared (LWIR) and neuro-morphic (Event) camera technologies for real world and consumer electronic applications.
- Working towards building new multi-modality synthetic datasets using text-to-image and image-to-image and text-to-video transformations by employing stable diffusion and large language models.
- Deploying the neuromorphic and thermal imaging pipelines on edge devices including Nvidia Jetson Nano, Nvidia Jetson Xavier and Raspberry Pi 5 boards for real world automation tasks by performing model optimization using advanced model pruning and quantization methods like ONNX, Tensor RT and Tensorflow Lite.
- Exploring efficient image and video grounding techniques, prompt engineering and using large language models (LLM's) for robust training of customized vision language models.
- Further developed and deployed advanced generative AI models, including stable diffusion and few-shot learning approaches, for the synthesis of high-quality, domain-specific multimodality datasets in medical imaging, face biometrics, and real-world industrial automotive application. Demonstrated proficiency in integrating vision-language models for data augmentation and improving model robustness across various medical modalities.
- Managing a multidisciplinary team of Ph. D scholars, research associates, and postdoctoral researchers across various AI research projects.

September 2022 – September 2024, Postdoctoral Researcher, at University of Galway and Machine Learning Intern, at Tobii Galway

- During my Postdoctoral research I worked on DTIF-DAVID and ADAPT projects. The main objective is the development of a multimodal (sound and vision) AI processing platform with low cost and low power consumption to be used for consumer electronic applications.
- Working with various image modalities which includes thermal, RGBD, event neuromorphic and further integrating it with ML algorithms for building smart imaging pipelines for different computer vision applications.
- Working towards generating large-scale synthetic datasets using the composite structure of computer vision algorithms which includes Image-to-Image translation based Generative Adversarial Networks and Text-to-Image based stable diffusion models for optimal training and validation of deep learning networks.
- Optimizing the CNN networks using various quantization methods like ONNX, and TensorRT for deploying it on low powered edge devices for real time testing.
- Managing and working with other team members and fellow colleagues for producing effective research outcomes and further presenting the work at various forums.
- Focusing on deep learning model tuning, validation for deployment on single board low-powered edge devices for real-time data processing.
- Mentoring Ph.D. and master's students in their research projects by providing technical assistance and helping them in drafting publications and research-based ethical approvals.
- Served as Program Chair for the [25th Irish Machine Vision and Image Processing \(IMVIP23\)](#) Conference where I successfully set up the program committee, paper submission portal, decision announcement of full and short papers via Microsoft CMT in due time frames, designing the conference program and conducted a training workshop on Image synthesis and its potential usage for real-world applications.
- Moreover, I have been involved in designing the course content, teaching, and conducting labs of Mobile Devices, Embedded System design, and Analog Electronics modules for bachelor and master's students.

September 2019 – August 2022 Ph.D. and Machine Learning Researcher, Xperi Corporation and University of Galway

- Worked on industry based European Union (EU) funded '[HELIAUS](#)' project as a part of my Ph.D. research project/ work placement with overall funding of 485,873.75 Euros allocated for National University of Ireland Galway.
- Research and implementation of state-of-the-art algorithms from the domains of computer vision, machine learning, deep neural networks, and edge computing for implementation of smart in-cabin driver monitoring and road monitoring system (smart forward sensing perception system).
- Integrating various image modalities which includes thermal, visible (RGB) and depth data with machine learning algorithms to build AI-based artificially intelligent imaging pipelines.
- Data Acquisition, optimization, and deployment of trained machine learning (ML) networks on graphical processing unit (GPUs) and low-powered single board edge devices.
- During this project, I have worked and successfully deployed various in-cabin and out-cabin applications which include autonomous thermal gender classification, large-scale synthetic thermal data generation, face localization, and facial landmarks detection on thermal facial frames, object detection in thermal spectrum using SoA YOLO-v5 and YOLO-V8 framework, neural network optimization using TensorRT and Tensorflow-Lite tools and deploying the optimized network engines on resource-constrained edge-GPU devices which includes Nvidia Jetson Nano and Xavier development boards.
- Collaborating with a team of research engineers and professionals from different European companies which include Xperi Corporation based in Ireland, Lynred based in France, Next2U based in Italy, and Denso based in Germany working under Heliaus project as the consortium partners for effective cross-fertilization of ideas.
- Drafting and publishing the research findings at various international computer vision conferences and engineering Journals.
- Collaboration with other Ph.D. researchers for brainstorming on different ongoing projects which include human facial depth estimation using single 2D images, creating new large-scale synthetic datasets for computer vision applications, etc.

August 2016 - July 2019 Computer Vision Engineer, PNEC-NUST, Karachi-Pakistan

- Lead and provide in-depth research expertise in the domains of computer Vision, machine learning, deep neural networks, and graphical processing unit (GPU) Servers Management for an intelligent security system to be replaced with the conventional CCTV camera system for enhanced university campus security.
- Successfully integrated and deployed intelligent facial recognition and automatic number plate recognition system with conventional CCTV camera system on various entrance points of the university campus.
- Develop plans and strategies for ongoing research projects under the supervision of higher management to outline time & cost-based feasible methods to be followed as per university guidelines.
- Recommend process improvements and pitch new ideas to improve the efficiency and capabilities of existing systems.

- Writing research grants for large-scale projects focused on building safe and smarter cities using artificial intelligence (AI).
- Preparing presentations, and user guide manuals of different project modules for technical and non-technical staff.
- Drafting and publishing the research findings and experimental results at various international conferences.

Key Research Projects/ Publications

1. Advanced Thermal Perception Systems for Automotive AI

Focus: Driver monitoring, ADAS, multimodal sensing, edge deployment, open-source outdoor thermal LWIR dataset

- Developed AI thermal perception pipelines for object detection, classification, tracking, and facial analysis
- Deployed optimized models on NVIDIA Jetson edge devices, and Raspberry Pi Boards
- Applications: in-cabin driver monitoring, out-cabin all environment object detection, releasing open-source datasets

Selected Publications:

1. "Evaluation of Thermal Imaging on Embedded GPU Platforms for Application in Vehicular Assistance Systems" published in *IEEE Transactions on Intelligent Vehicles*, doi: 10.1109/TIV.2022.3158094.
Authors: Muhammad Ali Farooq, Waseem Shariff, and Peter Corcoran
IF: 14.3, [\[Paper\]](#), [\[Github\]](#)
2. "Object Detection in Thermal Spectrum for Advanced Driver-Assistance Systems (ADAS)," published in *IEEE Access*, doi: 10.1109/ACCESS.2021.3129150.
Authors: Muhammad Ali Farooq, Peter Corcoran, Cosmin Rotariu and Waseem. Shariff
IF: 3.9, [\[Paper\]](#), [\[Github\]](#)
3. "Performance estimation of the state-of-the-art convolution neural networks for thermal images-based gender classification system," published in *SPIE Journal of Electronic Imaging*, doi: 10.1117/1.JEI.29.6.063004.
Authors: Muhammad Ali Farooq, Hossein Javidnia, Peter Corcoran
IF: 1.1, [\[Paper\]](#), [\[Github\]](#)
4. "On the Role of Thermal Imaging in Automotive Applications: A Critical Review," published in *IEEE Access*, doi: 10.1109/ACCESS.2023.3255110.
Authors: Muhammad Ali Farooq, Waseem Shariff, David O'Callaghan, A. Merla and Peter Corcoran
IF: 3.9, [\[Paper\]](#)
5. "C3I Thermal Automotive Dataset", available via IEEE Dataport, Muhammad Ali Farooq, Waseem Shariff, Faisal Khan, Peter Corcoran, Cosmin Rotariu, March 26, 2022.
[\[Dataset\]](#)

2. Generative AI & Synthetic Imaging Systems

Focus: Diffusion models, GANs, multimodal synthetic datasets

- Developed Stable Diffusion and GAN-based pipelines for synthetic data generation
- Built datasets across face biometrics, medical imaging, and thermal imaging
- Applied vision-language models and prompt engineering using LLM's

Selected Publications:

1. DiffusionTBAD: Rendering CTA images for type B aortic dissection diagnosis, published in *Computerized Medical Imaging and Graphics*, <https://doi.org/10.1016/j.compmedimag.2026.102740>.
Authors: Ayman Abaid, Muhammad Ali Farooq, Niamh Hynes, Peter Corcoran, Ihsan Ullah
IF: 4.9 [\[Paper\]](#)
2. "SynAdult: Multimodal Synthetic Adult Dataset Generation via Diffusion Models and Neuromorphic Event Simulation for Critical Biometric Applications," published in *IEEE Access*, doi: 10.1109/ACCESS.2025.3594875.
Authors: Muhammad Ali Farooq, Paul Kielty, Wang Yao, and Peter Corcoran
IF: 3.9, [\[Paper\]](#), [\[Github\]](#), [\[Project Website\]](#), [\[Dataset\]](#)
3. "ChildDiffusion: ChildDiffusion: Unlocking the Potential of Generative AI and Controllable Augmentations for Child Facial Data Using Stable Diffusion and Large Language Models" published in *IEEE Access*, doi: 10.1109/ACCESS.2025.3575964.
Authors: Muhammad Ali Farooq, Wang Yao, and Peter Corcoran
IF: 3.9, [\[Paper\]](#), [\[Github\]](#), [\[Project Website\]](#), [\[Dataset\]](#)
4. "ChildGAN: Large Scale Synthetic Child Facial Data Using Domain Adaptation in StyleGAN" published in *IEEE Access*, doi: 10.1109/ACCESS.2023.3321149.
Authors: Muhammad Ali Farooq, Wang Yao, Gabriel Costache, and Peter Corcoran
IF: 3.9, [\[Paper\]](#), [\[Github\]](#), [\[Project Website\]](#), [\[Dataset\]](#)
5. "ThermVision: Exploring FLUX for Synthesizing Hyper-Realistic Thermal Face Data and Animations via Image to Video Translation ", published in *Proceedings of the 33rd ACM International Conference on Multimedia (ACM-MM)*, 2025.
Authors: Muhammad Ali Farooq, Waseem Shariff, Peter Corcoran
[\[Paper\]](#), [\[Project Webpage\]](#), [\[Dataset\]](#)

6. "Enhancing Synthetic Image Realism with Controlled Diffusion Models ", published in IEEE 2025 International Joint Conference on Neural Networks (IJCNN), 2025.
Authors: Iqra Nosheen, [Muhammad Ali Farooq](#), Peter Corcoran, Cathy Ennis, Michael G. Madden
[\[Paper\]](#), [\[GitHub\]](#)

3. AI for Medical Imaging & Diagnostics

Focus: Skin cancer, CTA imaging, multimodal medical AI, thermography

- Developed AI systems for lesion detection, skin and breast cancer classification, and human thermography for disease prognosis
- Applied deep learning for cardiac signal estimation and vascular imaging

Selected Publications:

1. "A Comparative Study on Diffusion Sampling Methods Across Diverse Medical Imaging Modalities ", published in 17th Asian Conference on Computer Vision, (ACCV), 2024.
Authors: [Muhammad Ali Farooq](#), Ayman Abaid, Ihsan Ullah, Peter Corcoran
[\[Paper\]](#), [\[GitHub\]](#).
2. "Derm-T2IM: Harnessing Synthetic Skin Lesion Data via Stable Diffusion Models for Enhanced Skin Disease Classification using ViT and CNN", published in IEEE Engineering in Medicine and Biology Society (EMBC) 2024 conference.
Authors: [Muhammad Ali Farooq](#), Wang Yao, Michael Schukat, Mark A Little, Peter Corcoran
[\[Paper\]](#), [\[Github\]](#), [\[Dataset\]](#)
3. "Synthesizing CTA Image Data for Type-B Aortic Dissection using Stable Diffusion Models", published in IEEE Engineering in Medicine and Biology Society (EMBC) 2024 conference.
Authors: Ayman Abaid, [Muhammad Ali Farooq](#), Ihsan Ullah, Peter Corcoran
[\[Paper\]](#)
4. "Visual Cardiac Signal Classifiers: A Deep Learning Classification Approach for Heart Signal Estimation from Video", published in *IEEE Access*, doi: 10.1109/ACCESS.2024.3472508.
Authors: Mohammad Moustafa, [Muhammad Ali Farooq](#), Amr Elrasad, Joseph Lemley and Peter Corcoran.
IF: 3.9, [\[Paper\]](#)
5. "Infrared Imaging for Human Thermography and Breast Tumor Classification using Thermal Images," published in 31st IEEE Irish Signals and Systems Conference (ISSC), 2020.
Authors: [Muhammad Ali Farooq](#) and Peter Corcoran
[\[Paper\]](#)
6. "Automatic lesion detection system (ALDS) for skin cancer classification using SVM and neural classifiers." published in IEEE 16th International Conference on Bioinformatics and Bioengineering (BIBE). IEEE, 2016.
Authors: [Muhammad Ali Farooq](#), Muhammad Aatif Mobeen Azhar, and Rana Hammad Raza
[\[Paper\]](#)

4. Industrial Automation & Intelligent Vision Systems

Focus: Smart surveillance, industrial inspection, automation

- Developed AI-based surveillance systems (face recognition, ANPR)
- Applied deep learning for industrial quality inspection

Selected Publications:

1. "Synthetic Face Ageing: Evaluation, Analysis and Facilitation of Age-Robust Facial Recognition Algorithms," published in *IEEE Transactions on Biometrics, Behavior, and Identity Science*, doi: 10.1109/TBIOM.2025.3536622
Authors: Wang Yao, [Muhammad Ali Farooq](#), Joe Lemley and Peter Corcoran
IF: 5, [\[Paper\]](#)
2. "Automatic Inspection of Seal Integrity in Sterile Barrier Packaging: a Deep Learning Approach," published in *IEEE Access*, doi: 10.1109/ACCESS.2023.3348779.
Authors: Julio Z. Diaz, [Muhammad Ali Farooq](#), and Peter Corcoran
IF: 3.9, [\[Paper\]](#)
3. "A Study on the Effect of Ageing in Facial Authentication and the Utility of Data Augmentation to Reduce Performance Bias Across Age Groups," published in *IEEE Access*, doi: 10.1109/ACCESS.2023.3312612.
Authors: Wang Yao, [Muhammad Ali Farooq](#), Joe Lemley, and Peter Corcoran
IF: 3.9, [\[Paper\]](#)
4. "Effectiveness of State-of-the-Art Super Resolution Algorithms in Surveillance Environment", published in 8th Machine Intelligence and Digital Interaction Conference (MIDI), 2020.
Authors: [Muhammad Ali Farooq](#), Ammar Ali Khan, Ansar Ahmed, and Rana Hammad Raza

5. Neuromorphic & Efficient Edge AI Systems

Focus: Event-based vision, low-power AI

- Designed low-power multimodal AI systems for real-time applications

- Optimized models using TensorRT, ONNX, TensorFlow Lite

Selected Publications:

1. "Optimization of Event Camera Bias Settings for a Neuromorphic Driver Monitoring System". published in *IEEE Access*, doi: 10.1109/ACCESS.2024.3371487.
Authors: Mehdi Sefidgar Dilmaghani, Waseem Shariff, Muhammad Ali Farooq, Joe Lemley, Peter Corcoran
IF: 3.9, [\[Paper\]](#)
2. "Neuromorphic Driver Monitoring Systems: A Computationally Efficient Proof-of-Concept for Driver Distraction Detection," published in *IEEE Open Journal of Vehicular Technology*, doi: 10.1109/OJVT.2023.3325656.
Authors: Waseem Shariff, Mehdi S, Paul Kielty, Joe Lemley, Muhammad Ali Farooq., Peter Corcoran
IF: 4.8, [\[Paper\]](#)
3. "Development, optimization, and deployment of thermal forward vision systems for advance vehicular applications on edge devices" published in SPIE Fifteenth International Conference on Machine Vision (ICMV) 2022.
Authors: Muhammad Ali Farooq, Waseem Shariff, Faisal Khan, Peter Corcoran
[\[Paper\]](#), [\[Github\]](#)
4. "Event-based YOLO object detection: proof of concept for forward perception system" published in SPIE Fifteenth International Conference on Machine Vision (ICMV) 2022.
Authors: Waseem Shariff, Muhammad Ali Farooq, Peter Corcoran
[\[Paper\]](#)

Funding & Research Leadership

- Active contributor to EU Horizon 2020, SFI, DTIF, Wellcome Career Development, and ADAPT projects
- Experience in proposal writing, project planning, and stakeholder collaboration
- Collaborations with industry partners (Xperi, Lynred, Denso, Tobii, FotoNation, Boston Scientific)

Skills

Technical Skills

1. IP Algorithms: Segmentation, Filter Design, Noise Removal, Compression, Super Resolution
2. Imaging Modalities: Thermal (LWIR and NIR), Neuromorphic (Event data), Medical (Dermatoscopic, CTA, XRAY), RGB and RGBD
3. Object Det: SSD, YOLO-NAS, YOLOX, YOLO World, Faster R-CNN, DETR, GroundingDINO, YOLOv8
4. OCR docTR, PaddleOCR, EasyOCR, Tesseract OCR, MmOCR
5. DL techniques: Tracking, Optical Flow, Particle Filtering, Pose Estimation
6. CV Algorithms: Image Classification and Detection, Tracking
7. Network optimization: ONNX, TensorRT, TensorflowLite, OpenVINO
7. Programming Lang: Python, C, C++, Unix Shell Scripting
8. Deployment: Docker and Kubernetes
9. Editor: Visual Studio Code, PyCharm, Jupyter Notebook
10. OS: Ubuntu 18.04, Ubuntu 20.04, Redhat, Windows Server 2016, Linux
11. APIs: PyTorch, TensorFlow, Caffe, OpenCV, Flask, Gradio
12. Robotics/ Simulators: CircuitLab, Proteus, Matlab, Simulink, AutoCAD
13. Electronics: Digital Circuit Desing, Analog Electronics, PLC's, Industrial Electronics

Management Skills: Project management, team player, time management, effective communication, determination.

Honors and Achievements

- Serving as Senior Member IEEE (Membership no: 99096332)
- Serving as the Guest Editor for MDPI Electronics Special Issue titled "Advances in Computer Vision and Smart Imaging Technologies"
- Won the prestigious European Union (EU) H2020 Ph.D. scholarship.
- Active reviewer in various prestigious computer vision conferences and top engineering journals which includes IEEE ACM-Multimedia, IEEE International Joint Conference on Neural Networks (IJCNN), IEEE Transactions on Intelligent Vehicles (IEEE-TIV), IEEE Transactions on Image Processing (IEEE-TIP), IEEE Access, SPIE Journal of Electronic Imaging (SPIE-JEI), SPIE Journal of Applied Remote Sensing (SPIE-JARS), Nature Scientific Reports.
- Awarded NUIG employability award, lifts leadership award, and Alive presidential award for being an active member as an event manager in the University Postgraduate society, Seas Suas volunteer, and student union class representative.
- Awarded various technical certifications which include Convolution Neural Networks by Andrew NG, Research Integrity (Engineering and Technology) from Oxford University Press, IT security awareness and protecting PII by KnowBe4, and reviewer certifications from Nature and SPIE Journal bodies.